



Army University of Science and Technology (BAUST), Saidpur
Department of Computer Science and Engineering (CSE)
Midterm Examination, Winter/Summer 2025

B.Sc. Engg. in CSE
01 (One) hour

Course Code: CSE 3209 (Syl. 2021)

Credit: 3.00

Level- 3 Term- II

Full Marks: 45

Course Title: Information System Design

- Symbols and abbreviations bear their standard meanings.
- Assume reasonable values for any missing data.

(Answer any three questions including Question No. 1)

- | | Marks | COs/
CLOs |
|--|-------|--------------|
| 1. a) Sara, a CSE student, conducted an interview with Mr. Rahim Khan, the manager of a reputable software company for her academic purpose. Mr. Rahim discussed his professional journey and the company's efforts during the interview. He discussed the difficulties of project management and technology adaptation, emphasizing the value of diligence, collaboration, and lifelong learning. Sara was inspired by the interview to maintain her concentration and confidently pursue her career ambitions in technology.

i) Explain the proper steps that Sara must follow to make the interview successful.
ii) List and explain the skills that Mr. Rahim Khan must possess for handling project management and technology adaptation. | (15) | 1 |
| 2. a) Define System. Briefly explain the keyword 'suprasystem'. | (05) | |
| b) Illustrate the characteristics of an open systems. | (10) | |
| 3. a) A new learning app was developed by a TechNova Solutions software team. Before making the program available to users, they carefully designed it in collaboration with the customer to understand their needs and made sure it functioned as intended. Following the launch, they kept an eye on the app, addressed bugs, and adjusted in response to user input.

i) Using the correct terminology for each step, describe and explain the various phases that the software team followed.
ii) Describe the possible causes or conditions that could lead to a project's termination. | (15) | |
| 4. a) Give a brief description of the procedures that go into project planning. | (05) | |
| b) List and explain the tools used to gather information. | (10) | |



Bangladesh Army University of Science and Technology (BAUST)

Department of Computer Science and Engineering (CSE)

CT 2, Summer 2025

Level-3

Term-II

Course Code: CSE 3209

Course Title: Information System Design

Time: 25 Minutes

Date: 22/09/25

Full Marks: 15

Answer All Questions

Marks

1. Explain the essential skills required for an analyst to perform the job effectively.
2. List and briefly explain the sources and tools used for gathering information.

10

05



Bangladesh Army University of Science and Technology (BAUST)

Department of Computer Science and Engineering (CSE)

CT 3, Summer 2025

Level-3

Term-II

Course Code: CSE 3209

Course Title: Information System Design

Time: 25 Minutes

Date: 30/10/25

Full Marks: 15

Answer All Questions

Marks

05

10

1. List the steps of feasibility analysis.

2. Suppose, BAUST is thinking about spending 1,30,000tk on a microcomputer system to improve its documentation for safe storage monitoring. Over the course of its four-year operation, the system is expected to produce an average monthly benefit of 2,645tk with the interest of 1.5%.

Now, calculate the total benefit for the given situation to analyze whether the scenario profitable or not.



Bangladesh Army University of Science and Technology (BAUST)

Department of Computer Science and Engineering (CSE)

CT 1, Summer 2025

Level-3

Term-II

Course Code: CSE 3209

Course Title: Information System Design

Time: 25 Minutes

Date: 28/08/25

Full Marks: 15

Answer All Questions

Marks

1. Define System. Briefly explain the keyword 'suprasystem'.
2. Describe the characteristics of a system.

5

10

Bangladesh Army University of Science and Technology (BAUST), Saidpur
Department of Computer Science and Engineering (CSE)
Midterm Examination, Summer 2025

Program: B.Sc. Engg. in CSE Course Code: CSE 3201 (Syl. 2021)
Time: 01 (One) hour Credit: 3.00

Level-3 Term-II
Full Marks: 45

Course Title: Artificial Intelligence

- N.B. • Symbols and abbreviations bear their standard meanings.
• Assume reasonable values for any missing data.

(Answer any three questions, including Question No. 1)

1. a) Mr. Zubair is a Software Engineer who has been developing a chess game that is powered by AI. He created two special characters, Junior and Senior. Each can move to the left, right, up, or down, but cannot move diagonally. Both of them have complete knowledge of the game space and the legal moves. After he had successfully developed the game, he analyzed that 70% of the moves that the junior makes are attacking. But the senior attacks and defends when it is necessary.

Marks (07)
COs/
CLOs 1

(i) Define AI agents with a proper diagram and find out the rational agent of this game with justification.

- b) You are given a set of chromosomes of the form $x = abcdefgh$, where each a to h represents a gene value between 0 and 9. The fitness value of this chromosome can be calculated as follows:

(08) 3

$$f(x) = (a + b + c + d) - (e + f + g + h)$$

The chromosome lists are given below:

$$x_1 = 2\ 3\ 9\ 2\ 1\ 2\ 8\ 5$$

$$x_2 = 8\ 7\ 1\ 2\ 6\ 6\ 0\ 1$$

$$x_3 = 6\ 5\ 4\ 1\ 3\ 5\ 3\ 2$$

- (i) Evaluate the fitness value of each chromosome and arrange them in order with the fittest first.
(ii) Cross the fittest two individuals using one-point crossover at the middle point

2. a) Consider the following graph presented in Figure 2(a). Choose an algorithm between Breadth First Search(BFS) and Depth First Search(DFS) that would require the minimum number of operations to reach node MAX from node A. Justify your answer with a proper explanation.

(10)

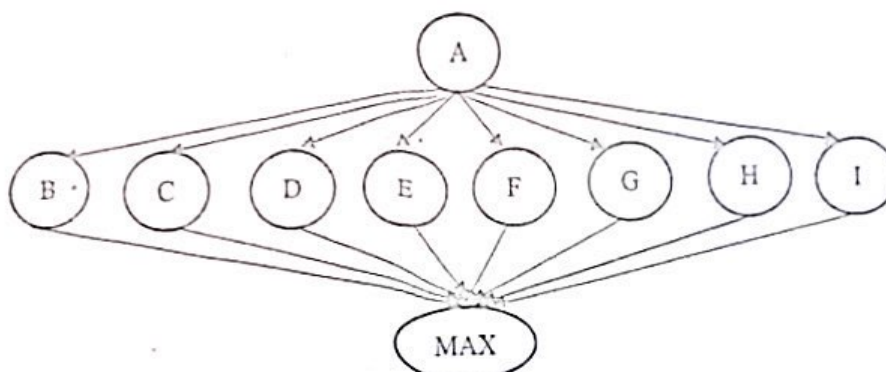


Figure 2(a): Directed Graph

- c) Define PEAS. Write down the PEAS for the BAUST website.

(05)

3. a) Consider the following graph, depicted in Figure 3(a). Perform the A* search algorithm on this graph to reach node G from node A. The heuristic values are provided in Table 3(a). (10)

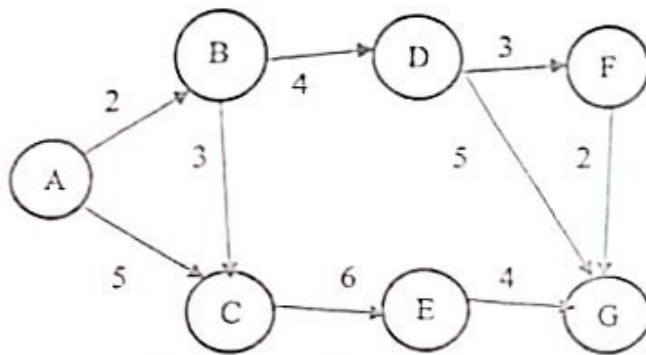


Figure 3(a): Directed Graph

Table 3(a): Heuristic value

A	10
B	11
C	9
D	7
E	8
F	6
G	0

- b) Convert the following statement to First-Order Predicate logic (FOPL). (05)
 "Every student loves exams."
4. a) You are given a Minimax game tree in Figure 4(a). (12)
- Apply the minimax algorithm and find the maximum value that MAX can gain.
 - Prove that alpha-beta pruning doesn't affect the minimax final result.

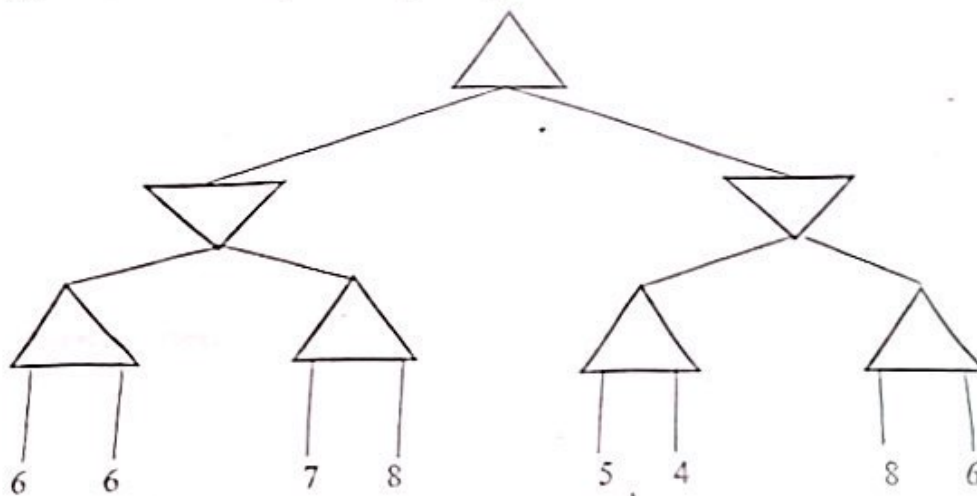
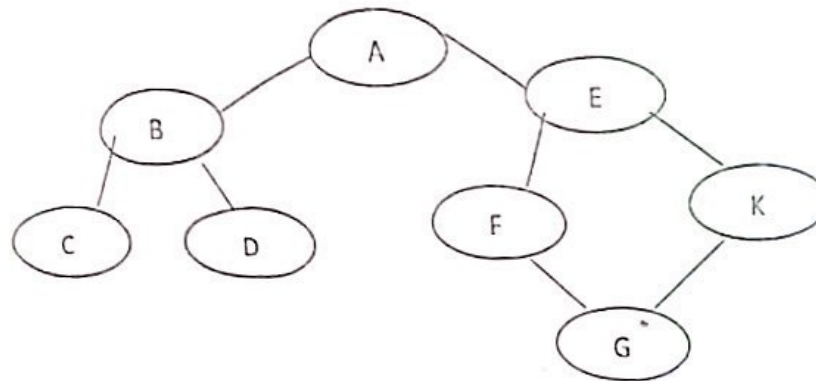


Figure 4(a): Game tree

- b) If $P(x)$ denotes $x > -1$, Find the truth values for the following statements: (03)
- $P(3) \wedge P(-2)$
 - $P(3) \rightarrow P(-1)$
 - $P(-4) \vee P(y)$

Topics: IDDS, 8-puzzle, ucs

(a) For the following graph in which iteration the node (G) will be visited by applying IDDS?



From	To	cost
A	E	20
A	B	15
B	C	10
B	D	25
E	F	35
E	K	10
F	G	30
K	G	10

(b) By applying UCS find the smallest path from A to G

(c) Solve the following 8 puzzle problem using A* search.

2	3	4
6		5
1	8	7

Initial State

1	2	3
4	5	6
7	8	

Goal State

Bangladesh Army University of Science and Technology (BAUST), Saidpur
Department of Computer Science and Engineering (CSE)

Course Code: CSE 3201 Course Title: Artificial Intelligence

CT 03, Summer 2025 Full Marks: 20 Time: 30min

Each question is worth 10 marks

- 1 A doctor knows that pneumonia causes a fever 95% of the time. She knows that if a person is selected randomly from the population, there is a 10^{-7} chance of the person having pneumonia. 1 in 100 people suffer from fever. You go to the doctor complaining about the symptom of having a fever (evidence). Find out the probability that pneumonia is the cause of this symptom (hypothesis) from the description below.

2

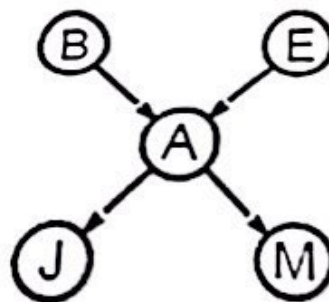
B	E	P(A)
true	true	0.95
true	false	0.94
false	true	0.29
false	false	0.001

A	P(J)
true	0.90
false	0.05

A	P(M)
true	0.70
false	0.01

$$P(B) = .001$$

$$P(E) = .002$$



Here B for Burglary, E for Earthquake, J for John, and M for Mary.
John calls (J) and Mary calls (M). What is the probability that neither burglary nor earthquake occurred if the alarm rang?



Bangladesh Army University of Science and Technology (BAUST)

Department of Computer Science and Engineering (CSE)

Class Test #3, Summer 2025

Level-3 Term-II

Course Code: CSE 3205

Course Title: Computer Networks

Time: 25

Date: 20/11/25

Set A

Full Marks: 15

Minutes

Answer All Questions

CLOs

Marks

1. Explain 'Optimality Principle' with an example.
2. Determine the shortest path and the corresponding minimum total weight from the starting node 'C' to the node 'D'.

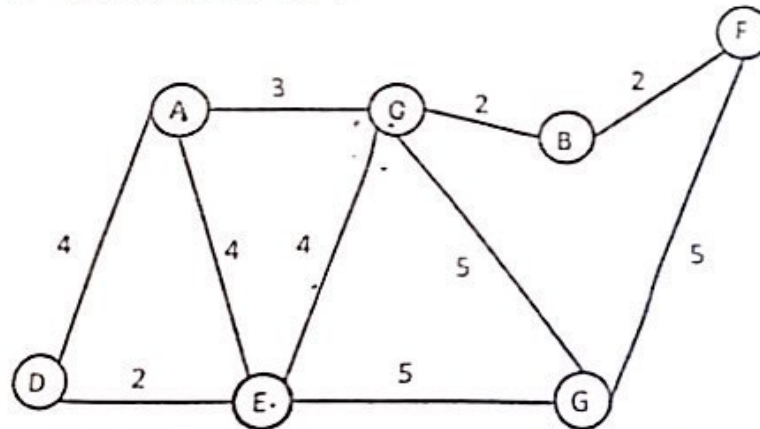


Figure 2: Weighted graph representation of a network



Bangladesh Army University of Science and Technology (BAUST)

Department of Computer Science and Engineering (CSE)

Class Test #2, Summer 2025

Level-3 Term-II

Course Code: CSE 3205

Course Title: Computer Networks

Time: 25

Date: 09/11/25

Full Marks: 15

Minutes

Answer All Questions

CLOs

Marks

1. Compare the Leaky Bucket and Token Bucket algorithms in terms of their working principles and packet handling mechanisms.
2. Draw a neat and properly labeled diagram of the TCP header. Explain the function and significance of each flag field included in the header structure.

2

06

2

09



Bangladesh Army University of Science and Technology (BAUST)

Department of Computer Science and Engineering (CSE)

Class Test #1, Summer 2025

Level- 3 Term II

Course Code: CSE 3205

Course Title: Computer Networks

Time: 25

Date: 14/09/25

Full Marks: 15

Minutes

Answer All Questions

CLOs Mars

1. A company is granted the site address 192.168.1.0 and the company needs 8 subnets. Design the subnets (Mention network address, valid host range and broadcast address). CLO2 10
2. Summarize the task of Application layer and Presentation layer of OSI model CLO1 5

- N.B. • Symbols and abbreviations bear their standard meanings.
 • Assume reasonable values for any missing data.

(Answer any three questions including Question No. 1)

	Marks	COs/ CLOs
1. a) Explain the different types of errors that can occur during data transmission in computer networks, providing a proper example for each.	(06)	2
b) A company is assigned the network address 190.160.4.0/24. Perform VLSM subnetting for the network in figure 1(b) and for each subnet, specify: i) Network address ii) Broadcast address iii) First usable IP address iv) Last usable IP address v) Number of valid hosts	(09)	3

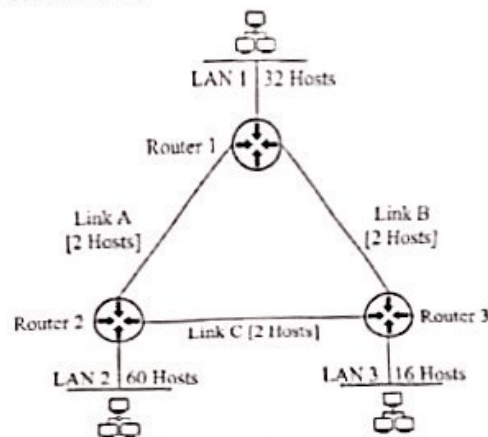


Figure 1(b): Network Architecture of the company

2. a) Compare and contrast Pure ALOHA and Slotted ALOHA. Illustrate your answer with suitable diagrams and provide justification for the superiority of one over the other. (08)
- b) Explain the functions of each layer of the TCP/IP model in detail, with appropriate examples. (07)
3. a) Explain the working principle of the Simplex Stop-and-Wait Protocol for a noisy channel with the help of a suitable diagram. (08)
- b) Describe Circuit Switching, Packet Switching and Message Switching with suitable diagrams and distinguish among them on the basis of key characteristics. (07)
4. a) Station A is required to send a message of 14 packets to Station B using a sliding window protocol with a window size of 4 and the Go-Back-N error control strategy. All packets are available for immediate transmission. If every 6th packet transmitted by Station A is lost (while no ACKs from Station B are lost), determine the packet transmission sequence at the sender side. (07)
- b) Using the Cyclic Redundancy Check (CRC) error detection method: (08)
 - i) Compute the transmitted codeword for the binary message 10110011 when the generator polynomial is $x^3 + 1$. Show all intermediate steps of the binary division.
 - ii) At the receiver side, assume the received codeword is 10110011101. Verify whether the received data is error-free. Justify your answer with proper calculation.



Bangladesh Army University of Science and Technology, Saidpur
Department of Computer Science & Engineering
CT-02, Summer 2025

Course Code: CSE 3207

Time: 25 mins

Level-3, Term-II (B)

Full Marks: 15

Course Title: Mathematical Analysis for Computer Science

	Marks	CLOS
1. For the same values of n and k , which produces a larger result, the Stirling number of the first kind or the second kind? Explain your answer with an example, where n and k retain their usual combinatorial meanings.	5	01
2. Find the general integer solution of the linear Diophantine equation: $6x + 9y = 5$	4	01
3. An electrical system consists of four components. The system works if components A and B work and either of the components C or D works. The reliability (probability of working) of each component is A: 0.9 B: 0.9 C: 0.8 and D: 0.8 . Find the probability that, (a) The entire system works. (b) The component C does not work, given that the entire system works. Assume that the four components work independently.	6	01

Bangladesh Army University of Science and Technology (BAUST), Saidpur
Department of Computer Science and Engineering (CSE)
Midterm Examination, Summer 2025/ Winter 2025

Program: B.Sc. Engg. in CSE Course Code: CSE 3207 (Syl 2021)

Level- 3 Term- II

Time: 01 (One) hour

Credit: 3.00

Full Marks: 45

Course Title: Mathematical Analysis for Computer Science

- N.B. • Symbols and abbreviations bear their standard meanings.
 • Assume reasonable values for any missing data

(Answer any three questions including Question No. 1)

	Marks	COs/ CLOs
1. a) Write the recursive formula for the Tower of Hanoi problem when two auxiliary (helper) towers are available. Then, using that formula, calculate step by step the minimum number of moves needed to transfer 10 disks from the source tower to the destination tower using two auxiliary towers.	(05)	1
b) There is a famously clueless group of people called the "Biddas". They are known in town for being very large in number but knowing almost nothing about mathematics. One day they order a huge round pizza to feed everyone. Because there are so many Biddas, they want to cut the pizza into the maximum possible number of slices using n cuts. Being bad at math, the Biddas begin cutting the pizza using only straight cuts across it. You suggest that they use zig cuts instead so that the number of slices increases, but they refuse to believe you. As you are skilled in mathematics, demonstrate mathematically which cutting method (straight cuts vs zig cuts) produces the greater number of slices for the same number of cuts.	(05)	1
c) A recursive algorithm is an algorithm that solves a problem by calling itself on smaller subproblems until reaching a base case. The Josephus problem is a theoretical problem in which n people stand in a circle and are eliminated in steps of k (every k -th person is removed) until only one person remains. The goal is to determine the position of the last survivor. Here, n is the total number of people, and k is the step count used to eliminate people. Now write a recursive algorithm or pseudocode that takes n and k as inputs and returns the position of the last surviving person. (Assume that the position starts from 0)	(05)	3
2. a) Find generalized and delimited form of the given sequence i) $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ ii) $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$	(04)	
b) Prove that the closed formula of the recurrence $R_0 = \alpha$; $R_n = R_{n-1} + \beta + \gamma n$ is $R_n = \alpha + n\beta + \frac{n(n+1)}{2}\gamma$ Also using the formula calculate the value of $\sum_{k=0}^{200} (5 + 8k)$	(7+4)	
3. a) Write down the step-by-step procedure of Stein's algorithm (Binary GCD algorithm) to find the greatest common divisor (GCD) of two numbers, u and v .	(05)	
b) Calculate the greatest common divisor (GCD) of 36 and 60 using Stein's algorithm (Binary GCD algorithm). Show each step of the calculation clearly.	(05)	
c) Prove that $GCD(a, b) * LCM(a, b) = a * b$	(05)	
4. a) Define coprime numbers. Then, provide a mathematical proof showing that there are infinitely many prime numbers in the universe.	(2+4)	
b) Evaluate the equivalence class of 5 with respect to congruence modulo 3.	(03)	
c) Determine the remainder when 100^{100} is divided by 9 that is $(100^{100} \bmod 9)$. Show all the steps clearly.	(06)	



Bangladesh Army University of Science and Technology, Saidpur
Department of Computer Science & Engineering
CT-03, Summer 2025

Course Code: CSE 3207

Level-3, Term-II (B)

Time: 25 mins

Full Marks: 15

Course Title: Mathematical Analysis for Computer Science

- | | Marks | CLOS |
|--|-------|------|
| 1. Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable X having the probability density function | 5 | CO2 |
| $f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$ | | |
| (a) Verify that $f(x)$ is a density function. | | |
| (b) Find $P(0 < x \leq 1)$. | | |
| 2. Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find | 10 | CO2 |
| (a) the joint probability function $f(x, y)$ | | |
| (b) $P[(X, Y) \in A]$ where A is the region $\{(x, y) x + y \leq 1\}$. | | |



Bangladesh Army University of Science and Technology, Saidpur
Department of Computer Science & Engineering
CT-01, Summer 2025

Course Code: CSE 3207

Level-3, Term-II (B)

Time: 25 mins

Full Marks: 15

Course Title: Mathematical Analysis for Computer Science

	Marks	CLOS
1. Answer the following questions:		
a. What is the minimum number of moves required to transfer 15 disks from source tower to destination tower using two auxiliary towers?	6	01
b. What is the maximum number of regions we can get using 10 lines?		
c. In the Joshephus problem if there are 1024 people numbered from 1 to 1024 then which person would win?		
2. In the Joshephus problem $J(n)$ determines the survivor's number among n people numbered from 1 to n . Now using the recursive formulas of $J(n)$ calculate the value of $J(300)$.	4	01
3. Write the recursive pseudo code of Tower of Hanoi with double auxiliary and derive the recursive formula from the code.	5	01





Bangladesh Army University of Science and Technology
Department of Computer Science and Engineering

Lab Final, Summer 2025
 Course Code: CSE 3204
 Time: 90 Minutes

Level-3 Term-II
 Course Title: Operating System Sessional
 Full Marks: 30

Date:

Answer any three out of four questions

CLOs Ma

CLO2 1

1. You are required to implement the C-SCAN (Circular SCAN) disk scheduling algorithm in C++. The program should:
 - i. Take the initial head position and a list of disk requests (cylinder numbers) from the user.
 - ii. Schedule the requests using C-SCAN.
 - iii. Show the order in which the requests are served, and the total seek time.
 - iv. Use the following input:
 Number of requests: 8
 Requests: 176 79 34 60 92 11 41 114
 Initial head position: 50

Range: [0, 179]

2. You have to write a C++ program to simulate memory allocation using the Best Fit. The program should:

CLO2 10

- i. Take input for the number of memory blocks and their sizes.
- ii. Take input for the number of processes and their sizes.
- iii. Allocate each process to a suitable memory block.
- iv. Display how each process is allocated (or not allocated) and the remaining space in each block after allocation.
- v. Use the following input:
 Enter number of memory blocks: 6
 Enter block sizes: 90 300 120 500 50 400
 Enter number of processes: 5
 Enter process sizes: 45 115 410 120 200
 Output will look alike:

Process	Size	Allocation	Remaining
0	212	1	
1	417	4	
2	112	1	
3	426	-1	

3. Write a C++ program that implements the First In First Out (FIFO) page replacement algorithm following these steps:

10

- i. Take input for the total number of pages.
- ii. Take the following sequence as input for the page reference sequence:
 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1
- iii. Take 3 as input for the number of frames in main memory.
- iv. Apply the FIFO algorithm to find how many page faults occur and print the number.

4. Implement the Banker's algorithm by completing the following steps:

10

- i. Create the following MAX matrix taking user input:

Process A	B	C
P0	7	5
P1	3	2
P2	9	0
P3	2	2
P4	4	3

- ii. Create the following ALLOCATION matrix taking user input:

Process A	B	C
P0	0	1
P1	2	0
P2	3	0
P3	2	1
P4	0	0

- iii. Now calculate the NEED matrix using the previous two matrix.
- iv. Take [3, 3, 2] as input for the AVAILABLE resources.
- v. Finally apply Banker's algorithm to determine whether the system is in safe state. If yes, print the safe sequence.

Answer All Questions

CLOs Marks

1. You are required to implement the **Shortest Job First (SJF) CPU scheduling algorithm (Non-Preemptive)** for a set of processes with given **Arrival Time (AT)** and **Burst Time (BT)**.

CLO1 4+6

Process ID	AT	BT
P1	0	5
P2	2	1
P3	1	3
P4	3	2

Tasks:

- Create a **Gantt chart** along with the table containing Process ID, AT, BT, CT, TT & WT (CLO1).
- Implement **Non-Preemptive SJF with AT** in C/C++.

Your program should:

- Take the number of **processes**, **AT**, and **BT** as input.
- Calculate **CT**, **TT**, and **WT** for each process.

2. a) Write bash command to create the following file hierarchy with a single command:
folder1 → folder 2
b) In folder 2 create a file named main.sh and in main.sh write bash code for printing "Hello World"
c) Now give appropriate execution permission and run the code to show output.
d) The chmod command in Linux is used to change file permissions. Explain the two different methods of using the chmod command to assign permissions to a file — one using symbolic notation and the other using numeric notation. Give suitable examples for each method.
3. Write a Bash script that takes marks of 5 students as input and stores them in an array and displays each student's result as "Pass" if the mark is 40 or above, otherwise "Fail". Use a loop to iterate through the array and an if-else statement to check the result for each student.

CLO1 1+1+1+3

CLO1 4



Bangladesh Army University of Science and Technology (BAUST)

Department of Computer Science and Engineering (CSE)

Class Test #3, Summer 2025

Level-3 Term-II

Course Code: CSE 3203

Course Title: Operating System

Time: 25 Minutes Date: 27/10/25

Full Marks: 20

CLOs	Marks
------	-------

Answer All Questions

- | | | | |
|----|--|------|----|
| 1. | What is External fragmentation? How to solve external fragmentation? | CLO3 | 04 |
| 2. | Explain Belady's Anomaly. | CLO3 | 04 |
| 3. | Using Optimal Page Replacement Algorithm find the number of fault & fault ratio.
The reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5.
Number of free frames : 3 | CLO3 | 12 |



Bangladesh Army University of Science and Technology (BAUST)
Department of Computer Science and Engineering (CSE)

Class Test #2, Summer 2025

Course Code: CSE 3203

Time: 25 Minutes Date: 11/11/25

Level-3

Term-II

Course Title: Operating System

Full Marks: 20

CLOs

Marks

Answer All Questions

1. Define Synchronization and Critical Section.
2. Define Semaphore and write its two basic operations.
3. Explain the Dining Philosophers Problem.
 - Describe the problem scenario clearly.
 - Explain how it can be solved using semaphores.
 - Provide the code implementation.
 - Illustrate what happens when multiple processes (philosophers) arrive at the different time and when they arrive at same times.

CLO3

04

CLO3

04

CLO3

12

- S.R. • Symbols and abbreviations bear their standard meanings
 • Assume reasonable values for any missing data.

(Answer any three questions including Question No. 1)

**Marks COs/
CLOs**

- 1 a) Consider a system with 4 processes: P1 through P4 and 3 resources types: A (9 units), B (3 units), C (6 Units). Current allocation of resources and maximum requirement of a process for each resource are given in Figure for Question 1(a). Detect whether the current state of this system is safe or not.

(10) 3

Table 1(a): Process and Resource Information

Process	Allocation ABC	Max ABC
P1	100	322
P2	612	613
P3	211	314
P4	002	422

- b) Define operating system. Differentiate between timesharing and multiprogramming systems.

(05) 1

- 2 a) Consider the following workload

(15) 3

Table 2(a): Process Information

Process	Priority	Burst Time	Arrival Time
P1	3	70	40
P2	1	40	0
P3	1	100	10
P4	2	50	70

Draw the Gantt chart and calculate the average turnaround time and for each of the following scheduling algorithms:

- (i) Non-preemptive Shortest Job First
 (ii) Preemptive Priority Scheduling. Consider the processes enter FIFO queue according to their approval times. Within the same priority class, schedule according to FIFO queue.

3. a) Write down the circumstances under which CPU-scheduling decisions may take place. **(07)**

- b) Context switching how impacts on CPU utilization. **(08)**

4. a) Draw the resource graph for the following scenario where A, B, C, D, and E denote processes and 1, 2, 3, 4, and 5 denote resource types. There exists only one resource of each type. Show the steps of the execution of the deadlock detection algorithm on the constructed graph starting from node B. **(10)**

- (i) Process A holds 1 and 3, wants 2
 (ii) Process B holds 4, wants 3 and 5
 (iii) Process C holds nothing, wants 2
 (iv) Process D holds nothing, wants 2
 (v) Process E holds 5 and wants 1

- b) Describe the purpose of system call. State the reason that why system calls are not directly used. **(05)**